HARDWARE SPECIFICATION & BILL OF MATERIALS

VOTE-GUARD

First Working Hardware Prototype

Physical RFID Detection, Microcontroller Pinout & Sensory Feedback Architecture

[IMPLEMENTED: FRONTEND PROTOTYPE] [PLANNED / NEW: HARDWARE & SERIAL LINK]
[PROPOSED: PINOUT & SCHEMATICS] [FUTURE / OPTIONAL: OLED & CASING]
[TBD: BACKEND & DATABASE TECHNOLOGY]

DOCUMENT RELEASE
1.0.0 (Engineering Master)

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TARGET HARDWARE
ESP32 DevKit V1 (30-Pin) + RC522

TOTAL PROTOTYPE BUDGET
â‚¹850 â€“ â‚¹1,180 (~ â€“)

AUTHOR / ORGANIZATION
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DOCUMENT SCOPE
Hardware Requirements & BOM

Academic & Prototype Context: For the prototype, the RFID UID is used as a test identifier mapped to a voter record. UID-based identification alone should not be considered sufficient for secure real-world voter authentication. This prototype demonstrates physical RFID card detection, UID extraction, communication with a host terminal, voter record lookup, instantaneous duplicate-attempt interception, and physical sensory feedback. It does not provide production-grade election security.

Hardware Requirements & Bill of Materials (BOM)

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Project Scope: First Working Hardware Demonstration Prototype
Status: Hardware Planning & Component Specification
Author: VOTE-GUARD Engineering Team
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Classification Standard:
â€¢ `[IMPLEMENTED]` â€" Currently functional in the existing client-side HTML/CSS/JavaScript prototype
â€¢ `[PLANNED / NEW]` â€" Features intended for the physical hardware and connected integration phase
â€¢ `[PROPOSED]` â€" Conceptual hardware architectures, circuits, and schemas under evaluation
â€¢ `[FUTURE / OPTIONAL]` â€" Long-term extensions (OLED display, biometrics, custom casing, battery backup)
â€¢ `[TBD]` â€" Decisions reserved until further system review (backend framework, database engine)

ðŸ›¡ï¸ Official Prototype Scope & Academic Engineering Statement

VOTE-GUARD is currently a client-side voter verification and duplicate-attempt detection prototype.
The existing frontend prototype utilizes browser-based local storage (`localStorage`) and manual keyboard input to simulate voter verification.
The hardware design specified in this document defines the **First Working Hardware Prototype**. The physical prototype integrates an RFID reader and an embedded microcontroller to capture physical card test credentials and provide immediate sensory feedback (LEDs and buzzer).

- **College Prototype Context:** For the prototype, the RFID Unique Identifier (UID) is used as a test identifier mapped to a voter record. UID-based identification alone should not be considered sufficient for secure real-world voter authentication.
- **Scope of Demonstration:** The prototype demonstrates physical RFID card detection, UID extraction, communication with a host terminal, voter record lookup, instantaneous duplicate-attempt interception, and physical sensory feedback. It does **not** provide production-grade election security or official government voting services.
- **Backend Neutrality:** Backend technologies (Python, Node.js, Java, FastAPI, Spring Boot) and database engines (MySQL, PostgreSQL, MongoDB, SQLite) remain strictly `[TBD]` and are not assumed by this hardware specification.

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1. PROJECT HARDWARE OVERVIEW

VOTE-GUARD is designed to tackle one of the most critical vulnerabilities in electoral administration: **duplicate voting and voter impersonation at the polling booth**. While traditional paper registers rely on manual line-crossing and indelible ink, VOTE-GUARD demonstrates an automated, tamper-evident verification barrier.

The current working system exists as an interactive client-side web application `[IMPLEMENTED]`, where a polling officer manually enters voter credentials to simulate eligibility verification, status locking, and duplicate-attempt detection.

To bridge the gap between software simulation and real-world deployment, the project team has decided to prioritize building the **First Working Hardware Prototype** before selecting or implementing a centralized backend infrastructure. This hardware prototype introduces a physical RFID tap station that replaces manual typing with contact-free card reading and supplements onscreen dashboard alerts with unmistakable physical sensory feedback (illuminated LEDs and acoustic buzzer alarms).

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2. HARDWARE DESIGN PRINCIPLES & CONSTRAINTS

The development of the first physical prototype is guided by strict engineering and economic constraints:

1. **Backend Technology Independence** `[TBD]` **:**

The hardware prototype communicates with the host computer over standard universal serial streams. It does not presuppose whether the eventual backend will be written in Python, Java, Go, or Node.js, nor does it mandate a relational or document database.

1. **Current System Compatibility** `[IMPLEMENTED]` **:**

The hardware must easily map to the existing prototype data models (such as `RFID-88219`, `RFID-44012`, etc.) without breaking current dashboard workflows.

1. **Budget Accessibility for Academic Research:**

The Bill of Materials (BOM) must remain affordable for college engineering teams, keeping the total acquisition cost under â‚¹1,200 ($14.00) while using readily available commercial off-the-shelf (COTS) components.

1. **Electrical Safety & Reliability:**

All component voltages, logic thresholds, and current limits must be strictly verified. The design avoids fragile level-shifting hacks, respects ESP32 pin strapping constraints, and isolates inductive buzzer loads.

1. **No Scope Creep:**

Complex accessories such as graphic displays, motorized card locks, or biometric cameras are excluded from the initial prototype to ensure rapid, fail-safe assembly and reliable testing.

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3. VOTE-GUARD HARDWARE REQUIREMENTS

The physical prototype must satisfy five core functional hardware requirements:

- **HR-01: Proximity Card Detection:**

The station must automatically energize passive 13.56 MHz RFID transponders when placed within a 2 cm to 5 cm proximity field.

- **HR-02: Unique Identifier (UID) Extraction:**

The embedded controller must reliably read the transponder's factory-burned 4-byte or 7-byte UID and convert it to a standardized ASCII hex string.

- **HR-03: Workstation Serial Connectivity** `[PLANNED / NEW]` **:**

The controller must establish a bidirectional, plug-and-play USB Serial communication link (115200 baud) with the polling officer's workstation.

- **HR-04: Tri-State Sensory Feedback** `[PLANNED / NEW]` **:**

The hardware must provide immediate, distinct physical feedback for the three operational outcomes:

- **Valid Voter:** Solid Green LED + single affirmative confirmation tone.
- **Duplicate Voting Attempt:** Rapidly strobing Red LED + high-intensity triple alarm siren.
- **Unregistered Card:** Solid Yellow LED + double error buzz.
- **HR-05: Bus-Powered Simplicity:**

The entire hardware apparatus must operate solely from standard USB 5V bus power without requiring an external AC wall adapter.

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4. HARDWARE ARCHITECTURE

The hardware architecture organizes the prototype into three distinct physical domains: the **Polling Booth Hardware Station**, the **Communication Interface**, and the **Host Workstation / Terminal**.

FIGURE 01 **VOTE-GUARD Hardware System Architecture** [RECOMMENDED & PLANNED]

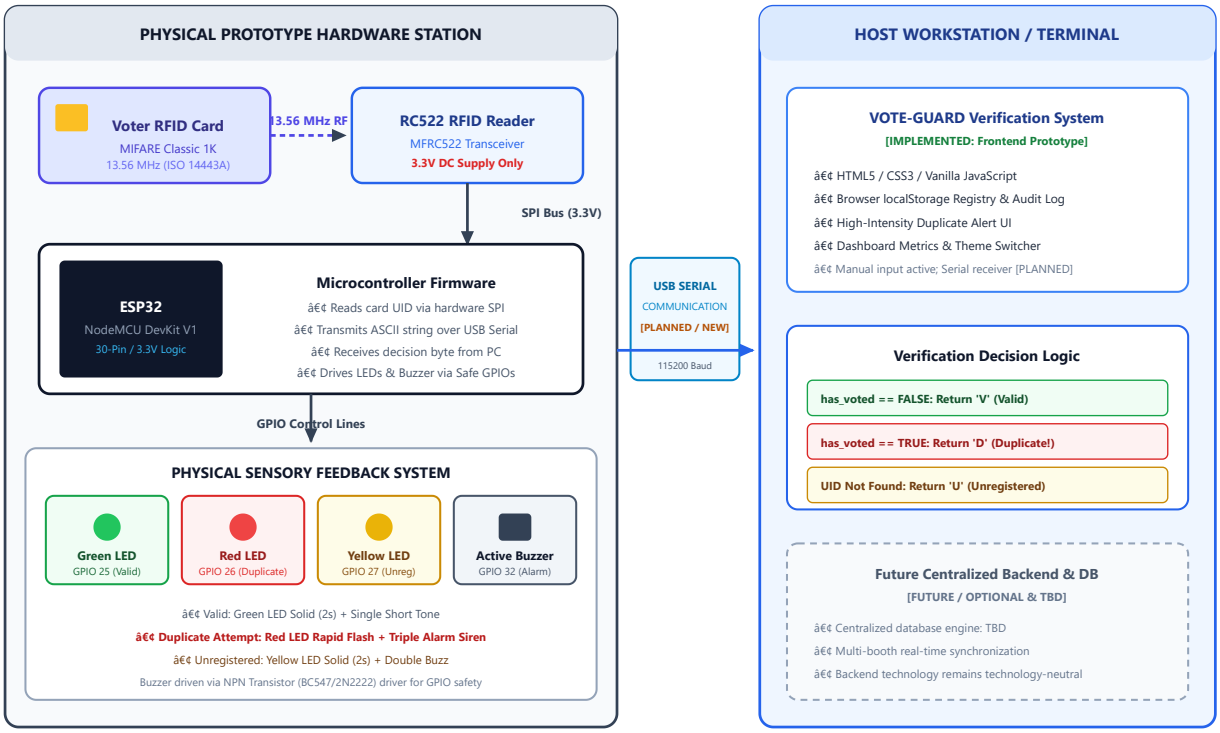


Figure 01: Block diagram showing RFID card coupling, ESP32 processing, USB serial link, host verification, and physical feedback.

5. HARDWARE COMPONENT CLASSIFICATION

Every item in the VOTE-GUARD hardware ecosystem is systematically classified to ensure focused procurement and clear milestone execution:

Classification	Definition	Hardware Scope
A. Essential Components	Minimal hardware required to scan physical RFID cards, extract their UIDs, send them to the workstation, and actuate physical sensory feedback.	ESP32 DevKit V1, RC522 RFID Module, MIFARE Classic Cards, Status LEDs (Green, Red, Yellow), Active Buzzer, NPN Transistor, Resistors, Breadboard, Jumper Wires, USB Cable.
B. Recommended Assembly Items	Consumables and items that enhance testing efficiency, component protection, and physical presentation.	Adhesive Card Labels (for tagging test voter names/IDs), Anti-static Sleeves, Acrylic mounting plate.
C. Optional / Future Hardware	Advanced components reserved for future multi-booth synchronization, offline edge autonomy, or exhibition casing.	0.96" I2C OLED Display, 5V 2A DC Wall Adapter, ABS Project Box Enclosure, Biometric Fingerprint Sensor, Thermal Receipt Audit Printer.

6. COMPLETE BILL OF MATERIALS (BOM)

The Bill of Materials below specifies all components required to construct the First Working Hardware Prototype.

No.	Component	Recommended Model / Type	Qty	Purpose	Status	Approx. Market Price (INR)	App Mar Price (USD
1	**Microcontroller Board**	**ESP32 NodeMCU DevKit V1** (30-pin, CP2102/CH340)	1	Master embedded controller; manages SPI RFID polling, USB serial bridge, and feedback GPIOs.	[BUY NOW]	â‚¹350 â€“ â‚¹480	$4.2 â€“ $5.8
2	**RFID Reader Module**	**RC522 (MFRC522)** 13.56 MHz RFID Kit	1	Transmits 13.56 MHz carrier wave to read ISO 14443A card UIDs.	[BUY NOW]	â‚¹140 â€“ â‚¹200	$1.7 â€“ $2.4
3	**RFID Smart Cards**	MIFARE Classic 1K (13.56 MHz, S50, White PVC)	5	Physical voter credentials mapped to registered voter records.	[BUY NOW]	â‚¹70 â€“ â‚¹100	$0.8 â€“ $1.2
4	**RFID Keyfobs**	MIFARE Classic 1K 13.56 MHz Keyfobs (Blue ABS)	3	Portable test tokens used to simulate duplicate and unregistered voters.	[BUY NOW]	â‚¹30 â€“ â‚¹50	$0.3 â€“ $0.6
5	**Green Status LED**	5mm Diffused Green LED	1	Visual indicator for **Valid Voter Verification**.	[BUY NOW]	â‚¹2 â€“ â‚¹5	$0.0 â€“ $0.1
6	**Red Status LED**	5mm Diffused Red LED	1	Visual warning for **Duplicate Voting Attempt Interception**.	[BUY NOW]	â‚¹2 â€“ â‚¹5	$0.0 â€“ $0.1

No.	Component	Recommended Model / Type	Qty	Purpose	Status	Approx. Market Price (INR)	App Mar Pric (USI
7	**Yellow Status LED**	5mm Diffused Yellow LED	1	Visual indicator for **Unregistered / Unknown RFID Card**.	[BUY NOW]	â‚¹2 â€“ â‚¹5	$0.0 â€“ $0.1
8	**Active Piezo Buzzer**	5V / 3.3V Active Buzzer (Continuous Tone)	1	Auditory alert siren for card scan, success tone, and fraud warning.	[BUY NOW]	â‚¹15 â€“ â‚¹30	$0.1 â€“ $0.4
9	**NPN Driver Transistor**	**BC547** or **2N2222** (TO-92 Package)	1	Isolates buzzer current draw from ESP32 GPIO to ensure electrical safety.	[BUY NOW]	â‚¹2 â€“ â‚¹5	$0.0 â€“ $0.1
10	**Base Resistor (Buzzer)**	1.0 kÎ©, 1/4W Carbon Film Resistor	1	Limits base current into NPN driver transistor.	[BUY NOW]	â‚¹1 â€“ â‚¹2	$0.0 â€“ $0.0
11	**LED Limiting Resistors**	220 Î© or 330 Î©, 1/4W Carbon Film Resistors	3	Protects ESP32 GPIOs and LEDs from excessive current.	[BUY NOW]	â‚¹5 â€“ â‚¹10	$0.0 â€“ $0.1
12	**Solderless Breadboard**	830-Point MB-102 Full Breadboard	1	Prototyping platform for plug-and-play solderless wiring.	[BUY NOW]	â‚¹80 â€“ â‚¹130	$1.0 â€“ $1.6

No.	Component	Recommended Model / Type	Qty	Purpose	Status	Approx. Market Price (INR)	App Mar Pric (USI
13	**Dupont Jumper Wires**	20cm Wires (10 Male-Female + 10 Male-Male)	1 Set	Interconnects RC522 header pins, breadboard rails, and ESP32.	**[BUY NOW]**	â‚¹60 â€“ â‚¹90	$0.7 â€“ $1.1
14	**USB Data Cable**	Micro-USB (or USB-C depending on ESP32 board)	1	Supplies 5V bus power and provides bidirectional USB-UART serial link.	**[BUY NOW]**	â‚¹50 â€“ â‚¹90	$0.6 â€“ $1.1
15	**Adhesive Card Labels**	Self-Adhesive White Paper Labels	1 Sheet	Physical labelling for test cards (`RFID-88219` , `DUPLICATE-TEST` , etc.).	**[BUY NOW]**	â‚¹10 â€“ â‚¹20	$0.1 â€“ $0.2

Price Disclaimer: All prices listed above represent approximate market prices from standard Indian electronics distributors and online retailers (Robocraze, Robu.in, Amazon.in) as of October 2026. Actual market prices may vary based on supplier, local taxes, board variant, and seller.

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7. RFID READER SELECTION

Selecting the appropriate RFID technology requires balancing protocol security, reading distance, logic-level compatibility, and system complexity.

7.1 Detailed RFID Technology Comparison

Evaluation Metric	RC522 (MFRC522)	PN532 NFC Module	RDM6300 (125 kHz)
Operating Frequency	**13.56 MHz (High Frequency)**	13.56 MHz (High Frequency / NFC)	125 kHz (Low Frequency)
Standard Supported	**ISO/IEC 14443 Type A**	ISO 14443A/B, FeliCa, NFC P2P	EM4100 / TK4100
Credential Types	MIFARE Classic 1K/4K, Ultralight, S50 fobs	MIFARE, NTAG, Smartphone NFC Emulation	125 kHz Read-Only ID Tags
Host Interface	**Hardware SPI** (up to 10 Mbit/s), I2C, UART	I2C, SPI, High-Speed UART (HSU)	UART TTL (9600 baud)
Operating Voltage	**3.3V DC (Strict Requirement)**	3.3V â€“ 5.0V DC (selectable)	5.0V DC
Reading Range	**~2.0 cm â€“ 4.5 cm** (Controlled tap distance)	~2.5 cm â€“ 5.0 cm	~2.0 cm â€“ 6.0 cm
Software Support	Extensive (`MFRC522.h` by miguelbalboa)	Complex (`Adafruit_PN532`)	Basic serial stream parsing
Hardware Setup	Fixed pinout, straight SPI wiring	Requires onboard DIP switch bus selection	Fixed serial output
Module Cost (INR)	**â‚¹140 â€“ â‚¹200 ($1.70 â€“ $2.40)**	â‚¹350 â€“ â‚¹600 ($4.20 â€“ $7.20)	â‚¹150 â€“ â‚¹250 ($1.80 â€“ $3.00)
Suitability for Prototype	**Optimal** (Exact feature match)	Overkill (Added cost & configuration)	Poor (Outdated 125 kHz; unencrypted)

7.2 Technical Justification & Recommendation

The **MFRC522 (RC522)** reader module is chosen as the definitive RFID solution for the VOTE-GUARD First Working Hardware Prototype:

1. **Academic Prototype Suitability:** It is suitable for short-range RFID detection in a controlled prototype environment. The ~3 cm read zone mimics intentional voter card tapping while preventing accidental reads from neighboring

booths.

2. **Direct 3.3V Native Bus:** The RC522 operates natively at 3.3V DC, matching the ESP32 microcontroller logic level perfectly without requiring level-shifting resistors or bidirectional ICs.
3. **Firmware Maturity:** The standard `MFRC522` driver provides reliable anticollision routines, UID extraction, and interrupt management, enabling rapid development without low-level register debugging.
4. **Economic Feasibility:** At approximately â‚¹160, it provides maximum educational value per rupee.

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8. MICROCONTROLLER SELECTION

The microcontroller serves as the central hardware coordinator. It must handle high-speed SPI transactions with the RFID reader, maintain a robust USB serial link, and actuate status indicators without pin contention.

8.1 Controller Comparison Matrix

Feature / Criteria	**ESP32 NodeMCU DevKit V1**	**Arduino Uno (R3 / ATmega328P)**	**Arduino Nano (V3 / ATmega328P)**	**Raspberry Pi Pico (RP2040)**
Processor Architecture	32-bit Dual-Core Xtensa LX6	8-bit AVR RISC	8-bit AVR RISC	32-bit Dual-Core ARM Cortex-M0+
Clock Speed	**240 MHz**	16 MHz	16 MHz	133 MHz
Operating Logic Voltage	**3.3V DC Native**	5.0V DC	5.0V DC	3.3V DC Native
RC522 Compatibility	**Direct SPI Plug-and-Play**	Requires 5V-to-3.3V level shifters	Requires 5V-to-3.3V level shifters	Direct SPI Plug-and-Play
Hardware UARTs	3 Hardware Serial Ports	1 Hardware Serial (shared with USB)	1 Hardware Serial (shared with USB)	2 Hardware Serial Ports
Wireless Connectivity	**Built-in Wi-Fi 802.11 b/g/n & BLE**	None	None	None (Pico W required)
Available GPIO Pins	30 / 36 Pins (VSPI + HSPI)	14 Digital + 6 Analog	14 Digital + 8 Analog	26 Multi-function GPIOs
Form Factor	Breadboard-friendly DevKit	Large development board	Compact DIP-30	Compact DIP-40
Approximate Cost (INR)	**â‚¹350 â€“ â‚¹480 ($4.20 â€“ $5.80)**	â‚¹450 â€“ â‚¹700 ($5.40 â€“ $8.40)	â‚¹200 â€“ â‚¹320 ($2.40 â€“ $3.80)	â‚¹350 â€“ â‚¹450 ($4.20 â€“ $5.40)
Future Expansion Ready	**Yes (Cloud / IoT ready)**	Limited to USB tethering	Limited to USB tethering	Moderate

8.2 Recommendation: ESP32 NodeMCU DevKit V1 (30-Pin)

The **ESP32 DevKit V1 (30-pin version)** is selected as the recommended microcontroller for the prototype:

- **Elimination of Level Shifting:** Because the ESP32 operates natively at 3.3V, its SPI lines connect directly to the RC522. Connecting an Arduino Uno (5V logic) directly to an RC522 subjects the reader's input pins to 5V overvoltage, leading to heating, read errors, or permanent transceiver degradation.
- **USB-Serial Integration:** The onboard CP2102 or CH340 USB-to-UART bridge delivers rock-solid, bidirectional serial communication over a single USB cable.
- **Investment Protection for Future Milestones:** While the initial prototype relies on USB Serial, the ESP32 already possesses integrated Wi-Fi and Bluetooth. When the project progresses to multi-station wireless networking, the hardware will require **zero replacement**.

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9. PHYSICAL FEEDBACK SYSTEM

The sensory feedback system provides immediate, unambiguous physical confirmation at the polling station. It ensures that the voter and polling officer receive instantaneous sensory cues regardless of screen visibility.

9.1 Verification Outcome Signaling Matrix

Operational Event	Visual Feedback (LEDs)	Acoustic Feedback (Buzzer)	System Status & Meaning
Card Tap Detected	Green LED brief blink (30 ms)	Short click / chirp (30 ms)	RF antenna energized; card UID captured successfully.
1. Valid Voter (Eligible)	**Green LED Solid ON** (2.0 seconds)	**Single 150 ms affirmative beep** (clean tone)	Voter is registered and has not previously voted; ballot issuance authorized.
2. Duplicate Attempt (Fraud)	**Red LED Rapid Strobe** (50 ms ON / 50 ms OFF for 3.0 s)	**Three sharp high-intensity alarm beeps** (250 ms ON / 100 ms OFF Ã— 3)	**CRITICAL SECURITY ALERT:** Elector has already voted. Immediate booth halt; supervisor alerted.
3. Unregistered Card	**Yellow LED Solid ON** (2.0 seconds)	**Double error buzz** (200 ms ON / 100 ms OFF Ã— 2)	Credential UID not found in electoral roll; verification rejected.
4. Scan Error / Card Removed	All LEDs flash once (100 ms)	Single low buzz (300 ms)	Read corrupted or card pulled out prematurely; re-tap required.

9.2 Display Evaluation (Why No Display is Needed for Prototype 1)

- **OLED / LCD Screen Assessment:** An external display (such as a 0.96" I2C SSD1306 OLED or 16Ã—2 LCD) was evaluated but is **deliberately excluded from the First Prototype**.
- **Rationale:** The VOTE-GUARD host terminal already features a rich, responsive dashboard displaying voter photos, names, registration timestamps, live metrics, and high-visibility duplicate alert banners. Adding an auxiliary display increases BOM cost by ~â‚¹200, adds complex I2C wiring, draws additional current from the 3.3V rail, and duplicates data already presented on the operator's primary screen.

10. POWER SUPPLY & ELECTRICAL REQUIREMENTS

10.1 Power Architecture Overview

The prototype operates entirely from the host workstation's USB interface (+5.0V DC at up to 500 mA).

```
Host Computer USB Port (+5.0V DC, 500mA max)
       â”,
       â–¼ (Micro-USB / Type-C Data Cable)
ESP32 DevKit V1 Board
  â”œâ”€â”€ Microcontroller Core (+3.3V DC via on-board LDO Regulator)
  â”œâ”€â”€ 3.3V Power Rail Output Pin (3V3)
  â”,     â”""â”€â”€ RC522 RFID Reader VCC pin (~15â€“26 mA operating draw)
  â”œâ”€â”€ GPIO Control Outputs (3.3V Logic)
  â”,     â”œâ”€â”€ Green LED via 220Î© Resistor (~5.9 mA)
  â”,     â”œâ”€â”€ Red LED via 220Î© Resistor (~6.8 mA)
  â”,     â”œâ”€â”€ Yellow LED via 220Î© Resistor (~5.9 mA)
  â”,     â”""â”€â”€ NPN Base via 1kÎ© Resistor (~2.6 mA) â”€â”€> Switches Buzzer
  â”""â”€â”€ Common Ground (GND) Rail
```

10.2 Power Safety & Regulator Variance Guidelines

1. **USB Bus Power:** The complete prototype is powered via the USB connection to the host PC. No external power brick or lithium battery is needed for Prototype 1.
2. **Strict 3.3V Rail for RC522:** The RC522 module **MUST** be connected to the `3V3` power pin of the ESP32. Connecting RC522 VCC to the `VIN` or `5V` pin will destroy the MFRC522 IC.
3. **Common Ground:** All subsystems (ESP32, RC522, LEDs, buzzer driver) share a continuous, common electrical ground (GND) to maintain unified logic references.
4. **Regulator Variance Awareness:** Different commercial ESP32 DevKit boards utilize different Low Drop-Out (LDO) linear regulators (e.g., AMS1117-3.3, ME6211, or RT9013), with current limits varying between 250 mA and 800 mA. Because our complete peripheral load (RC522 + 1 active LED + transistor base) draws under 45 mA from the 3.3V rail, all standard DevKit regulators will operate well within safe thermal limits.

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11. ELECTRICAL COMPATIBILITY & SAFETY

11.1 ESP32 GPIO Selection & Boot-Strapping Pin Safety

The ESP32 features specialized multiplexed pins. Certain pins serve as **boot-strapping (strapping) pins** that dictate boot modes, flash voltage, and serial bootloader entry during power-on reset.

> [!CAUTION]
> **Boot-Strapping Pin Conflict Prevention:**
> - **GPIO 0:** Must be HIGH during normal boot. Attaching an external load can pull it LOW and force the ESP32 into ROM download mode.
> - **GPIO 2:** Must be left floating or pulled LOW during flashing. In many DevKits, it is hardwired to an onboard blue LED. Attaching external LED loads to GPIO 2 can disrupt boot stability.
> - **GPIO 12 & 15:** Control flash voltage (MTDI) and boot debug output.
> - **GPIO 34, 35, 36, 39:** These pins are **Input-Only (GPI)** and have no internal pull-up/pull-down or output drivers. They cannot drive LEDs or buzzers.
>
> **Resolution:** In this finalized design, **all strapping pins and input-only pins are strictly avoided**. The status indicators are assigned to dedicated, safe output GPIOs (**GPIO 25, 26, 27, and 32**).

11.2 Active Buzzer Electrical Safety Analysis

Standard 5V/3.3V active buzzers contain an internal electromagnetic coil and oscillation circuit. When energized, they draw between **20 mA and 35 mA**.

The ESP32 datasheet specifies a recommended continuous GPIO current limit of **12 mA** (with an absolute maximum rating of 40 mA). Driving a buzzer directly from an ESP32 GPIO can result in:

- Significant voltage sag on the 3.3V internal rail.
- Thermal stress on the microcontroller's internal output FETs.
- High-voltage inductive kickback (back-EMF) spikes when switched off.

Engineering Safety Solution:

The buzzer circuit is driven using a **low-cost NPN transistor switch (BC547 or 2N2222)**:

- **Base:** Connected to ESP32 **GPIO 32** through a **1.0 kÎ© current-limiting resistor** (draws only ~2.6 mA from the GPIO).
- **Collector:** Connected to the negative terminal of the active buzzer.
- **Emitter:** Connected directly to Common Ground (GND).
- **Buzzer Positive:** Connected to the ESP32 `VIN` (5V USB rail) or `3V3` rail.

(Note: If a pre-assembled 3-pin Active Buzzer Module is purchased, it already contains an onboard S8050 NPN transistor and base resistor on its PCB, allowing direct signal connection to GPIO 32).

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12. WIRING & PINOUT SPECIFICATION

FIGURE 02 **Physical Wiring & Pinout Architecture (ESP32 DevKit V1)** [SAFE HARDWARE PINOUT]

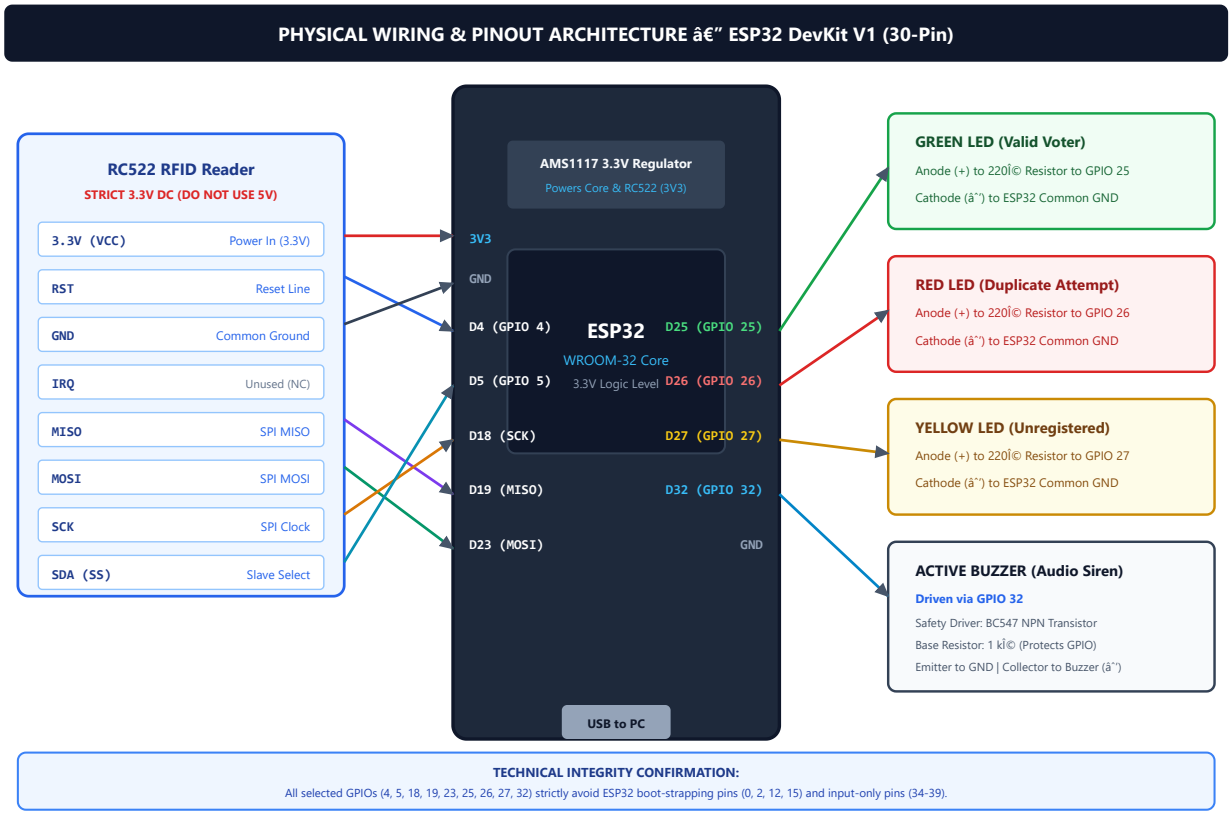


Figure 02: Detailed schematic mapping RC522 SPI lines, status LEDs with 220Î© resistors, and NPN-buffered active buzzer to safe ESP32 GPIOs.

Assumed Board Variant: ESP32 NodeMCU DevKit V1 (Standard 30-Pin Version).
If using a 36-pin or 38-pin ESP32 variant, verify the physical pin locations against the GPIO numbers listed below. The GPIO assignments remain identical in firmware.

12.1 Master Wiring Interconnection Table

Component Subsystem	Peripheral Terminal	ESP32 Board Pin	Assigned GPIO	Voltage / Logic Level	Functional Description
RFID Reader (RC522)	**3.3V (VCC)**	**3V3**	3.3V Power Rail	+3.3V DC (Regulated)	Main DC power for RFID module
	GND	**GND**	Common Ground	0V	Ground reference
	RST	**D4**	GPIO 4	3.3V Logic	Hardware Reset line for RC522
	MISO	**D19**	GPIO 19 (VSPI MISO)	3.3V Logic	SPI Master In / Slave Out (data to ESP32)
	MOSI	**D23**	GPIO 23 (VSPI MOSI)	3.3V Logic	SPI Master Out / Slave In (data from ESP32)
	SCK	**D18**	GPIO 18 (VSPI SCK)	3.3V Logic	SPI Synchronous Serial Clock
	SDA (SS / CS)	**D5**	GPIO 5 (VSPI SS)	3.3V Logic	SPI Chip Select (Slave Select)
	IRQ	*Unconnected*	â€”	â€”	Not connected (polled driver utilized)
Green Status LED	Anode (+)	Via 220Î© Resistor	**GPIO 25**	3.3V Logic	High = Valid Voter Verified
	Cathode (âˆ’)	**GND**	Common Ground	0V	Ground return
Red Status LED	Anode (+)	Via 220Î© Resistor	**GPIO 26**	3.3V Logic	High = Duplicate Attempt Alert
	Cathode (âˆ’)	**GND**	Common Ground	0V	Ground return
Yellow Status LED	Anode (+)	Via 220Î© Resistor	**GPIO 27**	3.3V Logic	High = Unregistered Voter ID
	Cathode (âˆ’)	**GND**	Common Ground	0V	Ground return
Active Buzzer	Positive (+)	**VIN** or **3V3**	Power Rail	+5.0V / +3.3V DC	Buzzer positive supply

Component Subsystem	Peripheral Terminal	ESP32 Board Pin	Assigned GPIO	Voltage / Logic Level	Functional Description
(via BC547 NPN)	Negative (â^')	Transistor Collector	â€”	Switched Ground	Switched path to ground
	Base Resistor	Via 1kÎ© Resistor	**GPIO 32**	3.3V Logic	High = Buzzer Sounding Active
	Transistor Emitter	**GND**	Common Ground	0V	Ground return
Host PC Link	Micro-USB Jack	USB Port	GPIO 1 (TX) / 3 (RX)	5V Bus / USB Serial	Power input and serial communication

13. HARDWARE-TO-SOFTWARE COMMUNICATION FLOW

The communication architecture links physical events at the polling booth to the VOTE-GUARD software running on the host workstation.

13.1 Protocol Specification `[PLANNED / NEW]`

- **Physical Layer:** USB 2.0 full-speed virtual serial port via CP2102/CH340 interface.
- **Baud Rate:** 115,200 baud, 8 data bits, no parity, 1 stop bit (8-N-1).
- **Upstream Message (Hardware to Computer):**

When an RFID card is tapped, the ESP32 transmits a clean, newline-terminated ASCII string:

`` `text

UID:8A4F123B\n

` ``

- **Downstream Command (Computer to Hardware):**

Following verification evaluation in the VOTE-GUARD software, the host returns a single ASCII command byte:

- `'V'` â€" **Valid:** Triggers Green LED for 2.0 seconds and single affirmative tone.
- `'D'` â€" **Duplicate:** Triggers flashing Red LED for 3.0 seconds and triple alarm siren.
- `'U'` â€" **Unregistered:** Triggers Yellow LED for 2.0 seconds and double error buzz.

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14. END-TO-END HARDWARE WORKFLOW

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+-----------------------------------------------------------------------------------------------
|                            STEP-BY-STEP SYSTEM OPERATIONAL WORKFLOW
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 1. SYSTEM INITIALIZATION & SELF-TEST
    â€¢ Hardware: ESP32 boots upon USB cable insertion, initializes SPI bus, resets RC522.
    â€¢ Hardware: Self-test routine momentarily flashes Green, Red, Yellow LEDs and sounds a 50
    â€¢ Software [IMPLEMENTED]: Officer launches VOTE-GUARD dashboard in workstation browser.
    â€¢ Software [PLANNED / NEW]: Web Serial connection opens COM port at 115200 baud.

 2. CARD PRESENTATION
    â€¢ Physical: Voter taps RFID smart card within 3 cm of the RC522 scanning antenna.

 3. RF COUPLING & UID EXTRACTION
    â€¢ Hardware: RC522 powers card via 13.56 MHz carrier, anticollision routine isolates UID.
    â€¢ Hardware: ESP32 captures 4-byte UID and sounds a momentary 30 ms chirp acknowledging de

 4. SERIAL TRANSMISSION
    â€¢ Hardware [PLANNED / NEW]: ESP32 formats UID and sends "UID:<HEX_STRING>\n" over USB Ser

 5. SOFTWARE RECEPTION & REGISTRY LOOKUP
    â€¢ Software [PLANNED / NEW]: Browser serial listener intercepts UID string.
    â€¢ Software [IMPLEMENTED]: Ingests UID, strips prefix, and executes verifyVoter(uid).
    â€¢ Software [IMPLEMENTED]: Queries voter registry (browser localStorage in current prototy

 6. ELIGIBILITY & DUPLICATE EVALUATION
    â€¢ Case A [VALID]: Voter registered AND has_voted == false.
      - Software: Marks has_voted = true, increments Recorded Count, renders green confirmation
    â€¢ Case B [DUPLICATE]: Voter registered AND has_voted == true.
      - Software: Increments Duplicate Count, logs incident, renders high-intensity red alert b
    â€¢ Case C [UNREGISTERED]: Credential UID not found in registry.
      - Software: Displays error notification ("Voter not found in registry").

 7. STATUS COMMAND DISPATCH
    â€¢ Software [PLANNED / NEW]: Sends feedback byte over USB Serial:
      - 'V' for Valid | 'D' for Duplicate | 'U' for Unregistered.

 8. PHYSICAL ACTUATION (HARDWARE)
    â€¢ Hardware [PLANNED / NEW]: ESP32 parses incoming byte:
      - If 'V': Green LED ON (2.0s) + single affirmative tone.
      - If 'D': Red LED rapid strobe (3.0s) + triple loud security alarm siren.
      - If 'U': Yellow LED ON (2.0s) + double error buzz.

 9. OFFICER VERIFICATION ACTION
    â€¢ Physical: Officer checks physical LEDs and dashboard confirmation.
    â€¢ Action: If Valid, issues ballot paper. If Duplicate, halts voting and flags security.

 10. STATION DEBOUNCE & IDLE RESET
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    â€¢ Hardware: ESP32 enforces a 1.5-second scan lockout to prevent re-reading the same card.
    â€¢ Hardware: LEDs extinguish; station returns to idle ready for the next voter.
```

15. PURCHASING CHECKLIST

This checklist isolates the components that must be procured to construct the First Working Hardware Prototype:

Item	Component Description	Recommended Qty	Priority	Approx Cost (INR)	Essential Purchasing Tips
1	ESP32 NodeMCU DevKit V1 (30-Pin)	1	**[BUY NOW]**	â‚¹380	Confirm pre-soldered header pins (avoids soldering).
2	RC522 13.56 MHz RFID Module Kit	1	**[BUY NOW]**	â‚¹160	Kit should include 1 card and 1 keyfob with straight pin header.
3	MIFARE Classic 1K Cards (13.56 MHz)	5	**[BUY NOW]**	â‚¹80	Ensure cards are 13.56 MHz (do **not** buy 125 kHz EM4100).
4	MIFARE Classic 1K Keyfobs (13.56 MHz)	3	**[BUY NOW]**	â‚¹40	Blue ABS keyfobs for dedicated duplicate/unregistered test cases.
5	5mm Diffused LEDs (1 Green, 1 Red, 1 Yellow)	3	**[BUY NOW]**	â‚¹10	Diffused casing provides superior viewing angles.
6	Active Buzzer (5V / 3.3V)	1	**[BUY NOW]**	â‚¹20	Verify it is **Active** (has sealed sticker on top; sounds with DC).
7	BC547 or 2N2222 NPN Transistor	1	**[BUY NOW]**	â‚¹3	Standard TO-92 general purpose NPN transistor.
8	Resistor Assortment (3Ã— 220Î©, 1Ã— 1kÎ©)	4	**[BUY NOW]**	â‚¹5	Standard 1/4 Watt carbon film resistors.
9	MB-102 Solderless Breadboard (830 points)	1	**[BUY NOW]**	â‚¹100	Full length breadboard fits ESP32 and all components cleanly.
10	Dupont Jumper Wires (10 M-F, 10 M-M)	1 Pack	**[BUY NOW]**	â‚¹70	20cm length provides comfortable spacing.
11	Micro-USB / Type-C Data Cable	1	**[BUY NOW]**	â‚¹60	Verify **Data Sync** capability (not charge-only).

Item	Component Description	Recommended Qty	Priority	Approx Cost (INR)	Essential Purchasing Tips
12	Adhesive Paper Labels	1 Sheet	[BUY NOW]	â‚¹10	For labelling test cards with voter IDs.
13	0.96" I2C OLED Display (SSD1306)	1	[OPTIONAL]	â‚¹200	Defer until initial prototype verification is complete.
14	ABS Project Box Enclosure	1	[FUTURE]	â‚¹150	Defer until final academic demonstration casing.

16. RECOMMENDED FIRST PROTOTYPE HARDWARE STACK

The final recommended hardware configuration for the VOTE-GUARD First Working Hardware Prototype comprises:

1. **Microcontroller: ESP32 NodeMCU DevKit V1 (30-Pin)**

Reason: Direct native 3.3V logic compatibility with RC522, integrated USB-UART serial engine, and built-in Wi-Fi for future backend expansion.

1. **RFID Reader: MFRC522 (RC522) 13.56 MHz RFID Reader Module**

Reason: Inexpensive, reliable short-range reading, direct 3.3V SPI operation, and industry-standard library support.

1. **RFID Test Credentials: MIFARE Classic 1K Cards (5 pcs) + Keyfobs (3 pcs)**

Reason: High-frequency 13.56 MHz passive credentials with unique factory UIDs, ideal for testing valid, duplicate, and unregistered scenarios.

1. **Green Status LED (5mm):**

Reason: Direct affirmative visual confirmation of an eligible, authorized voter.

1. **Red Status LED (5mm):**

Reason: High-intensity visual warning dedicated to intercepting duplicate voting attempts.

1. **Yellow Status LED (5mm):**

Reason: Unambiguous visual differentiation for unregistered or invalid credentials.

1. **Active Piezo Buzzer:**

Reason: Generates immediate acoustic alerts for fraud attempts without requiring complex frequency synthesis code.

1. **NPN Transistor Driver (BC547 / 2N2222) + 1kÎ© Resistor:**

Reason: Protects ESP32 GPIO from buzzer current draw and inductive switching spikes.

1. **LED Resistors (220Î©, 3 pcs):**

Reason: Limits forward current to safe levels (~5.9 mA), protecting ESP32 GPIOs and LEDs.

1. **Solderless Breadboard (830 points):**

Reason: Rapid, reliable, solder-free component assembly with easy multimeter probe access.

1. **Dupont Jumper Wires (10 M-F, 10 M-M):**

Reason: Flexible interconnection between the RC522 pin header, breadboard rails, and ESP32.

1. **USB Data Cable:**

Reason: Delivers single-cable 5V power and bidirectional Serial communication between the hardware and host PC.

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17. ESTIMATED PROTOTYPE COST

- **Estimated Cost (Indian Rupees): â‚¹850 â€“ â‚¹1,180**
- **Estimated Cost (US Dollars Equivalent): $10.20 â€“ $14.20**

> *Market Assessment: This budget represents the complete out-of-pocket procurement cost for all 15 essential items. The prototype remains extraordinarily cost-effective while delivering robust physical verification and fraud interception.*

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18. FUTURE HARDWARE EXPANSION `[FUTURE / OPTIONAL]`

As VOTE-GUARD evolves beyond the initial laboratory prototype into a connected election platform, the hardware architecture accommodates the following modular extensions:

1. **Dedicated Voter-Facing OLED Display:**

Connecting a 0.96" I2C SSD1306 OLED screen using the reserved I2C pins (**GPIO 21 SDA / GPIO 22 SCL**) to present voter guidance ("PLEASE TAP CARD", "VERIFIED").

1. **Wi-Fi / Ethernet Network Integration:**

Transitioning from tethered USB Serial to encrypted HTTPS/WebSockets using the ESP32's onboard Wi-Fi to sync with a centralized backend server.

1. **Multi-Factor Biometric Scanner:**

Adding an optical fingerprint sensor (e.g., R307 / AS608 via secondary hardware UART2 on GPIO 16/17) for two-factor voter verification.

1. **Battery Backup Power (UPS):**

Integrating a TP4056 lithium battery charging module and an 18650 cell to safeguard against polling station power interruptions.

1. **Tamper-Evident Physical Enclosure:**

A custom 3D-printed or laser-cut acrylic case featuring tamper-switch interlocks that log unauthorized opening attempts into audit storage.

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19. HARDWARE LIMITATIONS

To maintain academic rigor and transparent engineering documentation, the limitations of the first hardware prototype are explicitly documented:

1. **UID-Only Verification:**

The prototype reads only the factory-burned 4-byte or 7-byte UID. It does not utilize cryptographic key authentication (Crypto-1) or encrypted data blocks. Therefore, UID cloning is possible with specialized writable cards (UID changers/Chinese Magic Cards) in an untrusted environment.

1. **Tethered Workstation Operation:**

The prototype depends on an active USB cable connection to the host PC for power and serial data processing. It cannot operate as an autonomous standalone device.

1. **Single-Station Scope:**

Because the backend is not yet finalized, duplicate detection in Prototype 1 occurs within the host computer's local session. Cross-booth duplicate prevention requires the future centralized backend network.

1. **Prototyping Environment Fragility:**

Breadboard jumper connections are susceptible to accidental dislodging during demonstration handling compared to a custom Printed Circuit Board (PCB).

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20. CONCLUSION

The hardware specification and Bill of Materials outlined in this document establish a solid, practical, and highly cost-effective foundation for the **VOTE-GUARD First Working Hardware Prototype**.

By selecting the **ESP32 DevKit V1** and the **RC522 13.56 MHz RFID reader**, the system achieves direct 3.3V electrical compatibility, eliminates hazardous level shifting, and preserves native wireless capabilities for future milestones. The tri-state physical feedback system (Green, Red, Yellow LEDs + Active Buzzer with NPN isolation) provides unmistakable visual and acoustic fraud deterrence that directly mirrors the software dashboard's security alerts.

This hardware plan fulfills all project requirements while remaining strictly technology-neutral regarding future backend and database architectures. The procurement checklist and wiring definitions provided herein prepare the engineering team for immediate, error-free physical assembly.